

99th Dahlem Conference on Infection, Inflammation and Chronic Inflammatory Disorders: Psycho-neuroimmunology and the intestinal microbiota: clinical observations and basic mechanisms

This is a rapidly emerging field. The application of knowledge regarding the relationship between neural and immune systems in order to gain a better understanding of human conditions has been slow. In this discussion we describe how the brain and microbiota interact, and try to bring this into a context that is clinically relevant. We begin by describing established facts pertaining to the gut–brain axis and the role of gut bacteria. We then focus upon emerging data that will contribute to the generation of a new conceptual framework about the microbiota–gut–brain axis. In the final section we anticipate future directions of this field.